

RV-Sparse: CSR Sparse Matrix-Vector Multiply

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1 Overview

RV-Sparse implements sparse matrix-vector multiplication $y = A \times x$ using the Compressed Sparse Row (CSR) format, with zero dynamic memory allocation.

2 Implementation

The `sparse_multiply` function works in two phases:

1. **CSR construction:** Scan the dense row-major matrix A and extract non-zero values, column indices, and row pointers into caller-provided buffers.
2. **Multiply:** Compute $y_i = \sum_k values_k \times x_{col_indices_k}$ for each row i .

3 Repository

<https://github.com/vajradevam/RV-Sparse.git>